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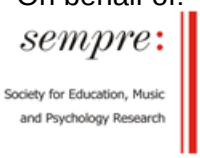
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Extramusical information contributes to emotions induced by music

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Abstract

Little is known about how extramusical, contextual information about a piece of music influences the emotions induced by that piece. The present study aimed to investigate this question by providing two groups of participants (both $n = 30$) with two different descriptions regarding the original context of a sad-sounding piece of film music; a description of a concentration camp scene or a description of a nature documentary. The results of these two groups were compared to previously collected data ($N = 60$), where participants listened to either the same, sad-sounding piece of music without a description, or to neutral-sounding music. The induced emotions were measured as objectively as possible using indirect memory and judgment tasks. The results suggest that contextual information about a musical piece may indeed influence the emotional effects of that piece, as the sad narrative description appeared to intensify the sadness induced by the sad-sounding piece. The narrative descriptions may have enhanced emotion induction via the *visual imagery* mechanism (suggested by Juslin & Västfjäll, 2008), as 80% of participants in both groups reported thinking about imagery related to the narrative descriptions provided.

Keywords

emotion, extramusical information, music, narrative, psychology

Music-induced emotions are not generated in a vacuum. Yet little is known about how extramusical factors such as contextual information influence the emotion elicitation process. Although instrumental music is able to effectively communicate affective (e.g., Gabrielsson, 2009) and semantic (e.g., Janata, 2004; Koelch et al., 2004) meaning without any explicit, extramusical information, one could argue that contextual information might significantly alter the interpretation of this meaning (see, e.g., Thompson, Russo, & Quinto, 2008) and thus influence the

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intensity and type of emotions induced. Simultaneously presented visual material has been found to influence the interpretation of emotions communicated by music (Thompson, Graham, & Russo, 2005; Thompson et al., 2008), but it is still unclear how prior, contextual information about a musical piece might contribute to the emotions induced by that piece.

Much of the music to which people are exposed in their daily lives – and have been exposed to throughout human history – such as songs, operas and film/television music, is intertwined with explicit narrative content. Even instrumental music without any apparent narrative context is often characterised in terms of stories or narratives (e.g., Tan & Kelly, 2004), possibly reflecting the frequent pairing of music and explicit narratives. It may also be that this tendency to 'hear stories' in music reflects something more innate. In fact, it has been suggested that the construction of narratives is fundamental to the way humans make sense of the world and their experiences in it (e.g., Polkinghorne, 1988). The idea of instrumental music conveying meaning in the form of implicit 'narratives' has been discussed quite extensively in the field of music philosophy (e.g., Cone, 1977; Maus, 1997; Newcomb, 1984; Treitler, 1989), but less so in the domain of music psychology (exceptions being Lavy, 2001; Shaffer, 1992). The view of the 'narrative' as an emergent property of the listening process (see Lavy, 2001), rather than an object or a property of the music itself, may hold the most promise for psychological studies on music and emotions. This 'narrative mode of listening' involves the detection and interpretation of cues from both sound and context (both circumstantial and musical), which become merged into one, coherent narrative structure (Lavy, 2001).

The idea of music listening as a narrative process is closely related to the phenomenon of music-induced visual imagery, which has been proposed as one of the mechanisms through which music can induce emotions in listeners (in addition to brain-stem reflexes, evaluative conditioning, emotional contagion, episodic memories, musical expectancy, and cognitive appraisal; see Juslin, Liljeström, Västfjäll, & Lundqvist, 2010; Juslin & Västfjäll, 2008). Music-induced visual imagery refers to a process whereby a listener conjures up – either intentionally or unintentionally – visual images while listening to music. These images may then – in combination with the music – produce an emotional response in the listener (Juslin & Västfjäll, 2008). Juslin and Västfjäll mention the possibility of a 'metaphorical nonverbal mapping between the music and so-called image-schemata grounded in bodily experience' (Juslin & Västfjäll, 2008, p. 566; see also Bonde, 2007), but the actual mechanisms through which music induces visual imagery are not yet clear.

We propose that music's ability to convey semantic meaning (see e.g., Koelch et al., 2004) and the human tendency to make sense of our experiences through the construction of narratives (see, e.g., Polkinghorne, 1988) is what gives rise to visual imagery in the context of music listening. It is still unclear to what extent music-induced visual imagery involves actual, pictoral representations of imagined events (cf. Juslin & Västfjäll, 2008), or whether the process would be better described as 'narrative imagery,' for example. In a recent experiment by Vuoskoski and Eerola (2012), a group of participants was asked to listen to 8 minutes of unfamiliar, instrumental, sad-sounding music and asked to freely describe the kinds of thoughts or impressions that went through their minds during the listening task. Of the participants, 43% reported thinking about imagery of some kind, such as 'The piece was quite melancholic and made me think of the ending of a sad movie. Maybe someone had died or gone away and left someone else missing him/her,' and 'The piece evoked impressions of nature; storm, water, wind, and swaying trees' (Vuoskoski & Eerola, 2012, p. 210). This suggests that spontaneous visual (or narrative) imagery may be present in a significant portion of music listening episodes – especially when music listening is the primary activity. Thus, it would be of significant interest to investigate the extent to which imagery influences music-induced emotions and to pinpoint factors that promote music-induced visual imagery.

The aim of this study is to investigate whether the emotions induced by a particular piece of music are influenced by prior information about the original context of that piece. We hypothesize that this explicit contextual information will have an effect on the emotions induced by the piece, possibly by promoting visual imagery related to its original context. We aim to investigate this by providing two groups of participants with two different descriptions regarding the original context of a sad-sounding piece of film music; a description of a concentration camp scene (*sad narrative*) or a description of a nature documentary (*neutral narrative*). In order to determine how the contextual information might alter the emotional effects of the music, the results of these two groups will be compared to previously collected (and previously reported; see Vuoskoski & Eerola, 2012) data, where participants listened to the same, sad-sounding piece of music without any contextual information. In addition, we will compare our results to a control group that listened to neutral-sounding music (data previously reported in Vuoskoski & Eerola, 2012), as this will provide a 'baseline' point of reference for any potential emotional effects. As suggested by Lavy (2001), the cues from the music and its context should become integrated into one, coherent narrative construct, which will be an integral part of the emotion-elicitation process. Thus it is hypothesized that the sad contextual information will intensify the sadness induced by the sad-sounding piece (as compared to the *no narrative* condition), while the neutral contextual information will attenuate it.

The induced emotions are measured as objectively and discreetly as possible using indirect memory and judgment tasks (for a review of indirect measures in the context of music and emotion studies, see Västfjäll, 2010), which also serve to minimize the effect of potential demand characteristics (see Orne, 1962). The rationale behind these indirect measures is based on the Associative Network Theory of Affect (Bower, 1981), according to which emotional states alter the processing of emotional information, causing affect-congruent biases. Previous studies have shown that induced affective states do indeed lead to affect-congruent judgment biases in the evaluation of emotional stimuli such as facial expressions and other types of pictures (e.g., Bouhuys, Bloem, & Groothuis,1995; Isen & Shalker, 1982). Thus people experiencing a sad emotional state should interpret emotional stimuli as sadder than people experiencing a neutral or happy emotional state (for a review, see Rusting, 1998). The Associative Network Theory also postulates that people should more easily remember material (such as words or memories) that is affectively congruent with their current emotional state. However, empirical evidence suggests that the opposite might be true (see, e.g., Isen, 1985; Parrot & Sabini, 1990; see also Vuoskoski & Eerola, 2012), and that people's instinctive attempts to regulate their moods and emotions lead to affect-incongruent memory effects instead – especially in the case of negative emotions (Isen, 1985). Parrot and Sabini (1990) demonstrated that affect-incongruent memory effects are observed when participants are unaware of the relevance of their affective states to the experiment, while affect-congruent memory effects emerge when participants are instructed to deliberately change their moods. Thus – as participants in the present experiment will not be informed about the relevance of their emotional states to the experiment – we expect that a sad emotional state should be associated with an affect-congruent judgment bias and an affect-incongruent memory bias.

Method

Participants

Participants were 60 Finnish university students from different faculties, aged 20–58 years (M = 25.1, SD = 5.4, 76.7% female). Of the participants, 58.4% identified themselves as

non-musicians, 33.3% reported doing music as a hobby, and 8.3% identified themselves as amateur or semi-professional musicians. Musical training was not a criterion in the recruitment of participants. Participants were randomly assigned to two conditions: A *sad narrative* condition or a *neutral narrative* condition. There were 30 participants in both conditions. In order to determine whether the narrative descriptions altered the emotions induced by the music, data from a previous experiment (conditions 1 and 2 from Vuoskoski & Eerola, 2012; $N = 60$) was included in some of the analyses for comparison purposes. The participants included in these data were aged 19–56 years ($M = 23.9$, $SD = 4.9$, 83.3% female).

Stimuli

Both groups listened to an excerpt (duration 8 minutes 35 seconds; starting at 2:25) from the instrumental piece *Discovery of the Camp* (Kamen, 2001, track 17). The piece has previously been validated as conveying strong sadness, whilst being unfamiliar to most listeners (for details, see Vuoskoski & Eerola, 2012). The two groups received different descriptions regarding the music. The *sad narrative* group received the following instructions (translated from Finnish by the first author), describing the original context of the piece:

> In the next section you will hear a piece of music (duration approx. 9 minutes) taken from *Band of Brothers*, a mini-series set in the Second World War. The music is taken from a scene where, in the final stages of the war, the Easy Company of the US Army's Airborne Division finds an abandoned concentration camp near Landsberg, Germany. The men of the Easy Company are deeply shocked by what they see; starving prisoners who have been left for dead. At first they don't understand the true purpose of the concentration camp, but they try their best to help the prisoners. Please concentrate on listening to the music – you can close your eyes if it helps you to concentrate. After the music ends, the experiment interface will automatically proceed to the next section.

In contrast, the *neutral narrative* group received the following instructions, describing a false yet plausible context for the music:

> In the next section you will hear a piece of music (duration approx. 9 minutes) taken from *Yellowstone*, a BBC documentary depicting the change of seasons and nature's awe-inspiring beauty in Yellowstone National Park. The music is taken from a section that illustrates how the rugged, impressive landscape of Yellowstone changes when autumn turns into winter. Please concentrate on listening to the music – you can close your eyes if it helps you to concentrate. After the music ends, the experiment interface will automatically proceed to the next section.

The data obtained from Vuoskoski and Eerola (2012) comprises two conditions (conditions 1 and 2 in Vuoskoski & Eerola, 2012) with similar tasks: In Condition 1, participants listened to the aforementioned music excerpt from the mini-series *Band of Brothers* without any narrative description provided. In Condition 2, participants listened to the first movement (*De l'aube à midi sur la mer*; duration 8 minutes 35 seconds) of the orchestral work *La Mer* by Claude Debussy. This piece has previously been validated as 'neutral' (i.e., neither positive nor negative) in terms of emotions expressed (for details, see Vuoskoski & Eerola, 2012). In the *Results and conclusions* sections of this article, these additional conditions will be referred to as *no narrative* and *comparison* conditions respectively.

Measures

Participants' affective state in the beginning of the experiment was measured using the *Positive and Negative Affect Schedule* (PANAS; Watson, Clark & Tellegen, 1988). The PANAS has 10 adjectives measuring positive affect and 10 adjectives measuring negative affect. Prior to the experiment, participants completed an empathy measure (*The Interpersonal Reactivity Index*; IRI; Davis, 1980). The IRI has 28 items (rated on a 5-point scale ranging from 0 to 4) that tap into four separate aspects of the global concept of empathy: Fantasy, Perspective-taking, Empathic Concern, and Personal Distress. The reason for investigating trait empathy in the context of the present experiment was that previous studies have associated empathy with susceptibility to music-induced sadness (Vuoskoski & Eerola, 2012) and with preference for sad music (Garrido & Schubert, 2011; Vuoskoski, Thompson, McIlwain, & Eerola, 2012). Participants also filled in a short background questionnaire about their music listening habits and musical expertise.

The emotions induced by the two music listening tasks were assessed indirectly using a word recall task and a picture judgment task. In the word recall task, participants had two minutes to write down as many adjectives as they remembered from the *Positive and Negative Affect Schedule* (PANAS) they completed in the beginning of the experiment (for a similar task, see e.g., Hartig, Nyberg, Nilsson, & Gärling, 1999). Note that participants were not instructed to memorize the PANAS adjectives. In the picture judgment task, participants rated the emotions expressed by 25 composite pictures depicting prototypical and ambiguous facial expressions (created by Vanger, Hoenlinger, & Haken, 1998). The five prototypical facial expressions used in the present experiment expressed joy, sadness, anger, fear, and neutrality, and the 20 ambiguous expressions were computer-generated combinations of these five prototypical expressions (for details, see Vanger et al., 1998).

Procedure

The procedure was almost identical to that described in the previous study by Vuoskoski and Eerola (2012) for conditions 1 and 2. A special computer program was designed in MAX/MSP graphical programming environment (running on Mac OS X) to run the experiment and to collect data. Participants were tested individually in separate rooms. In order to minimize the effect of demand characteristics, participants were told that the purpose of the study was to investigate the effect of music listening on cognitive processing, namely memory and perception. They were not informed about the role of emotions in these tasks. At the beginning of the experiment, participants rated their current affective state using the 20 adjectives of the PANAS (one adjective at a time). Half of the participants saw the adjectives in reversed order, so the order of the adjectives was balanced across participants. Participants were told that the purpose of the PANAS was to measure the extent of potential experiment fatigue. Next, participants listened to the instrumental sad music, accompanied either by a sad description (*sad narrative*) or a neutral description (*neutral narrative*). Participants listened to the music through studio quality headphones (AKG K141 Studio), and were able to adjust the sound volume according to their own preferences. After the music listening task, participants completed the word recall task described in the *Measures* section.

After 2 minutes had passed (during which participants wrote down as many adjectives as they recalled), the computer program moved automatically to the picture-judgment task. The 25 facial pictures were displayed in a different random order for each participant. Participants were instructed to rate the emotions each facial expression was portraying in their opinion

Table 1. Means and standard deviations (in brackets) for positive and negative mood prior to the experiment (PANAS), trait empathy scores, sadness ratings for facial pictures, and liking ratings for music stimuli.

	Sad narrative	Neutral narrative	No narrative	Comparison
Positive mood (before)	3.48 (0.70)	3.36 (0.47)	3.42 (0.70)	3.54 (0.56)
Negative mood (before)	1.27 (0.21)	1.33 (0.28)	1.47 (0.53)	1.44 (0.41)
Trait empathy (IRI)	72.33 (9.18)	68.93 (9.75)	71.30 (9.06)	70.40 (8.64)
Sadness ratings (*z* scores)	0.022 (0.17)	−0.027 (0.17)	0.000 (0.21)	−0.056 (0.20)
Liking ratings	5.73 (1.01)	5.23 (1.50)	5.33 (1.35)	4.60 (1.50)

using five rating scales (anger, fear, happiness, sadness and neutral) ranging from 1 (e.g., 'not at all angry') to 7 (e.g., 'very angry'). After the picture-judgment task, participants completed the PANAS questionnaire for the second time. Finally, participants were asked to describe what kinds of thoughts or impressions went through their minds during the music listening. They were also asked to rate how much they liked the music they heard (1 = not at all, 7 = very much). It took approximately 25 minutes to complete the experiment. After the experiment, all participants were fully debriefed and given a free movie ticket (value €14) in return for their participation. During debriefing, the participants in the *sad narrative* condition were asked whether they were familiar with the *Band of Brothers* mini-series and whether they recalled seeing the scene described in the narrative description (five participants recalled seeing the particular episode previously). Similarly, the participants in the *neutral narrative* condition were asked whether or not they had found the music familiar (none had found the music familiar).

Results

Mood prior to the experiment

In order to confirm whether any potential memory or judgment biases would reflect the hypothesized changes in participants' emotional states, one-way ANOVAs were carried out to analyse the PANAS ratings collected in the beginning of the experiment. In order to enable the fruitful comparison of results, the additional data from the previous study by Vuoskoski and Eerola (2012) was included in this analysis (*no narrative* and *comparison* conditions). There was no main effect of condition in the PANAS ratings, confirming that there were no existing mood differences between the groups prior to the music listening task. $F(3, 116) = 0.47$, $p = .71$, $\eta_P^2 = .01$ for positive affect, and $F(3, 116) = 1.94$, $p = .13$, $\eta_P^2 = .05$ for negative affect. The mean mood scores and standard deviations are displayed in Table 1.

Memory effects

The words were coded as correctly recalled if the root of the word was correct. For example, the words *pelokas* (*fearful*) and *peloissaan* (*afraid*) were regarded as equivalent. The mean numbers of correctly recalled positive and negative words (for all conditions) are displayed in Figure 1. The occurrence of potential memory biases (i.e., the number of correctly recalled positive words in relation to correctly recalled negative words) was investigated within conditions using one-tailed, paired *t* tests, as we had a clear hypothesis about the direction of the memory bias (cf. Parrot & Sabini, 1990, see also Vuoskoski & Eerola, 2012).[1] In the *sad narrative* condition, participants recalled significantly more positive words than negative words; $t(29) = 2.16$, $p = .020$,

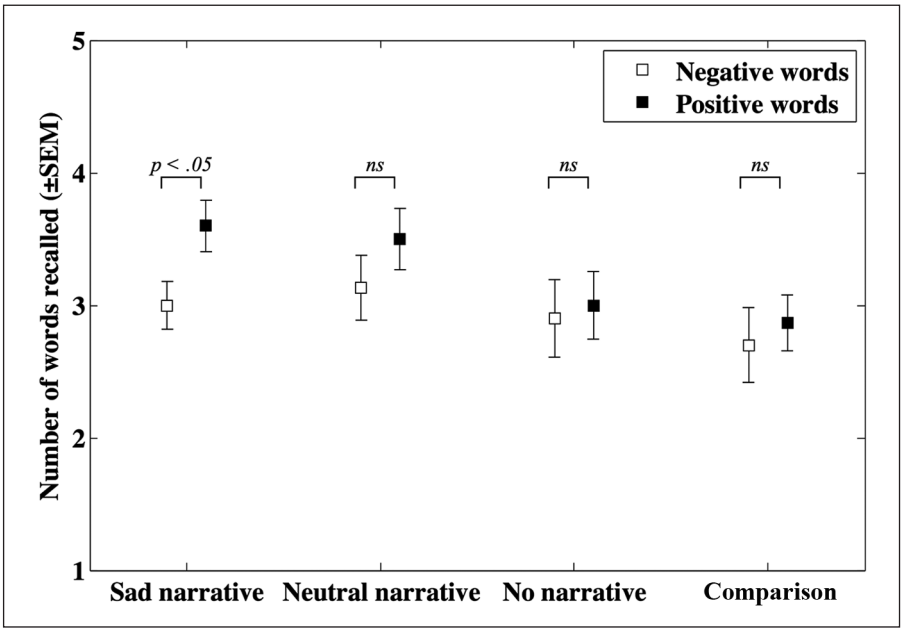


Figure 1. The mean number (± standard error of the mean) of correctly recalled positive and negative adjectives, grouped by condition

r^2 = .14, thus displaying an affect-incongruent memory bias. The participants in the *neutral narrative* condition displayed a similar – yet statistically non-significant – trend; $t(29) = 1.19$, $p = .12$, $r^2 = .05$. In comparison, there was no statistically significant difference between the number of correctly recalled positive and negative words in the *no narrative* and *comparison* conditions; $t(29) = 0.41$, $p = ns$, and $t(29) = 0.32$, $p = ns$, respectively.[2]

Judgment effects

To eliminate the effect of possible individual differences in scale use, the raw emotion ratings for the facial expression pictures were standardized within subjects using individual *z*-score transformations. The z-scores were calculated using all emotion ratings of each participant. According to the affect-congruency theory (e.g., Bower, 1981), people experiencing a sad emotional state should give higher sadness ratings to emotional – especially ambiguous – stimuli. To investigate whether the mean sadness ratings (calculated from each participant's sadness ratings for all facial expressions) differed between the conditions (*sad narrative, neutral narrative, no narrative*, and *comparison*), a 2-way ANOVA was run. The other between-subjects variable was level of empathy (high vs. low, median split), as the previous experiment by Vuoskoski and Eerola (2012) revealed that trait empathy played a significant role in the sadness judgments made by the participants who had listened to sad music (with no narrative provided). The median IRI score was 72, while the mean score was 70.74 (SD = 9.14). The ANOVA yielded no main effects of condition or level of empathy, but a significant interaction effect, $F(3,112) = 4.21$, $p = .007$, $\eta_P^2 = .10$. Post-hoc comparisons of means (Tukey) revealed that the level of empathy only made a difference in the *no narrative* condition, where the high

empathy participants gave significantly higher sadness ratings (to facial expressions) than the low empathy participants ($p = .027$).[3] It was further examined whether levels of trait empathy varied significantly between the four groups. A one-way ANOVA confirmed that there were no significant differences between the different groups with regard to trait empathy; $F(3,116) = 0.74$, $p = .53$, $\eta_P^2 = .02$. The mean sadness ratings and mean IRI scores for the different conditions are displayed in Table 1.

Free descriptions of thoughts that occurred during music listening

The participants' free descriptions about the kinds of thoughts and impressions that went through their minds while they were listening to the music were coded with regard to imagery related to the provided narratives. In the *sad narrative* condition, 24 participants (80%) mentioned thinking about imagery related to concentration camps and/or the Second World War (quotations translated from Finnish by the first author):

> I was thinking about the scene that was described, and I saw in my mind a camera panning in both directions, left and right, and seeing the sad faces of tortured people, and the horrified, sad expressions on the soldiers' faces. (Female, 22 years)

> Inspired by the description, my thoughts drifted to war and the atmosphere of a concentration camp. After the beginning, I also heard birds (swans) singing in my mind, and got the feeling of spring. In my mind, the war had ended, and people were gathering things from the ruins and starting to rebuild. Tired people, sacrifice, but there was some kind of hope. (Male, 21 years)

> I imagined how the soldiers arrive at the concentration camp, and find sick, starved people and dead people. Some of the people are crying and some are meeting their fates in silence. The soldiers are going back and forth, rushing to those who need help, or just look around in disbelief. I was very sad and almost cried when listening to the piece. (Female, 22 years)

Similarly, 24 participants (80%) in the *neutral narrative* condition reported thinking about nature-related imagery:

> In my mind, I had an image of a sloping forest next to a lake. I imagined a small cottage on the hillside, and myself standing in front of it. I imagined that the ground was still relatively bare, but snow was falling slowly from the sky. I felt peaceful and it felt good to just be and listen to the music. (Male, 23 years)

> The music accompanied me through different seasons, and I thought about the changes that happen in nature and about the feelings that those changes evoke. I imagined how water drops move differently in different seasons. (Female, 21 years)

The remaining 20% of the participants (in both groups) mostly reported thinking about things related to their daily lives (e.g., 'I was listening to the music feeling relaxed and with my eyes closed, and stuff related to my hobbies and friends came into my mind;' female, 25 years, *sad narrative* condition), although other types of imagery were also reported in some cases (e.g., '. . . while listening to the music, I imagined a situation where a person is leaving a loved-one behind. Another image that came into my mind was a withering flower;' Female, 25 years, *neutral narrative* condition).

Liking ratings

Finally, the liking ratings given to the music pieces were analyzed using a 4-by-2 factorial ANOVA, with Condition (*sad narrative, neutral narrative, no narrative*, or *comparison*) and Level of trait empathy (high vs. low) as the between-subject variables. Trait empathy was included in the analysis because previous studies have found a positive connection between the level of trait empathy and liking for sad music (e.g., Garrido & Schubert, 2011; Vuoskoski et al., 2012). The analysis revealed a main effect of Condition, $F(3,112) = 3.72$, $p = .014$, $\eta_P^2 = .09$ but no effect of Level of empathy and no interaction effects. Post-hoc analyses (Tukey) revealed that the participants in the *sad narrative* condition liked the music that they listened to significantly more than the participants in the *comparison* condition ($p = .008$). There were no statistically significant differences between the other groups. The mean liking ratings and standard deviations for the different conditions are displayed in Table 1.

Discussion

The results of this study suggest that contextual information about a musical piece can have an impact on the emotional effects of that piece. As hypothesized, the sad narrative description appeared to intensify the sadness induced by a sad-sounding piece of music, as the participants in this group displayed a significant memory bias in the word recall task. However, it appeared that the neutral narrative description did not attenuate the intensity of induced sadness as hypothesized, but the participants in the *neutral narrative* group actually displayed a non-significant trend in the same direction as the *sad narrative* group. It could be speculated that – when compared to the *no narrative* group – the visual imagery promoted by the narrative descriptions enhanced the participants' attention and immersion in the music and thus intensified the induction of emotion via music in both cases. However, the type of emotion induced may have been somewhat different in the two groups. These conclusions are indirectly supported by the finding that 80% of the participants in both groups reported thinking about imagery related to the descriptions given in the beginning of the music listening task. It appears that, as hypothesized, narrative descriptions about the original context of a piece of music do in fact promote visual imagery related to those descriptions, thus giving support to the view of visual imagery stemming from a narrative mode of listening.

Although the results of the picture judgment task remained largely inconclusive, they suggest that trait empathy only interacted with emotion induction in the *no narrative* condition. In the *no narrative* condition, empathic participants gave significantly higher sadness ratings to facial pictures than non-empathic participants. It was suggested previously (see Vuoskoski & Eerola, 2012) that this finding might have resulted from an interaction between trait empathy and the *emotional contagion* mechanism (see Juslin & Västfjäll, 2008). In other words, it was proposed that empathic people might be more susceptible to emotional contagion via music. However, it appears that the addition of a narrative description about the original context of the music attenuates this interaction. Although multiple emotion-induction mechanisms may be functioning simultaneously, it could be speculated that manipulations such as those employed in the present study can influence the primary mechanism of emotion induction. It is possible that in the present study the narrative descriptions promoted emotion induction via the *visual imagery* mechanism over other mechanisms. It is also possible that the narrative descriptions attenuated individual differences in emotional responding by restricting the free associations and spontaneous imagery otherwise evoked by the music.

The findings of the present study suggest that music can induce significant levels of sadness via the *visual imagery* mechanism, and that the content of imagery can be manipulated through extramusical information. The word recall bias in the *sad narrative* condition was even stronger than in a previously tested group listening to self-selected sadness-inducing music, which only displayed a non-significant memory bias (see Vuoskoski & Eerola, 2012). In this group, 60% of the participants reported thinking about autobiographical memories while listening to the music (cf. the *episodic memories* mechanism; Juslin & Västfjäll, 2008). It is also remarkable that the participants in the *sad narrative* condition seemed to like the sad music more than the other groups, although their liking ratings were only significantly different from the liking ratings of the *comparison* group who listened to neutral music. These findings are somewhat in contrast to those of Margulis (2010), who found that extramusical, dramatic, and structural descriptions (provided prior to music listening) reduced the enjoyment of musical pieces. Nevertheless, our results support previously made suggestions (see e.g., Vuoskoski & Eerola, 2012; Vuoskoski et al., 2012) that the enjoyment of sad music (and of sad fiction in general; Knobloch-Westerwick, Gong, Hagner, & Kerbeykian, 2012) may be positively associated with the intensity of sadness induced.

Although this study has provided preliminary evidence regarding how contextual information may affect music-induced emotions and promote visual imagery, there are some limitations that need to be discussed. First of all, this study only investigated one sad-sounding piece of music and how sad and neutral descriptions might influence the emotions induced in listeners. A fruitful approach for future studies might be to investigate more ambiguous music that is more sensitive to contextual influences, and thus offers more leeway for different interpretations. It may be that the neutral narrative (about autumn changing into winter in Yellowstone National Park) in the present experiment prompted melancholic connotations in some participants, reinforced by the sad music. However, it was deemed important that both stories were congruent with the music, as incongruent narrative descriptions would have likely been rejected by participants.

Although 80% of the participants in both groups reported thinking about imagery related to the provided narrative descriptions, we cannot know how intently they focused on the imagery, how much time they spent thinking about it, or whether this imagery would be better described as visual or narrative. We avoided giving specific instructions related to imagery (in order to minimize the effect of demand characteristics), but future studies should also try to investigate this aspect in more detail. Furthermore, there might have been other factors besides imagery that contributed to the increased intensity of felt sadness in the *sad narrative* condition, such as episodic memories (especially in the case of the five participants who recalled seeing the *Band of Brothers* episode previously), and the additive contribution of the narrative description as an affective stimulus (see e.g., Baumgartner, Esslen, & Jäncke, 2006). However, it is unlikely that the short narrative description could have had such a strong, lasting effect on participants' emotional states (as an independent stimulus) that this effect would have been visible and measurable 8.5 minutes after its presentation (studies using stories to induce affective states have tended to use considerably longer texts, and even then their emotional effect has been shown to be significantly reduced 3–4 minutes after their presentation; see e.g., Phillips, Bull, Adams, & Fraser, 2002). Instead, it is more likely that the music and the narrative interacted in a way that produced a stronger emotional response compared to the *no narrative* condition. We propose that this interaction is best explained as music-induced imagery, promoted by contextual information about the music.

Finally, the indirect measures of experienced emotion – although relatively free from problems such as demand characteristics and the need for accurate introspection – are not without

difficulties, as one has to rely on subtle and sensitive indirect effects such as memory and judgment biases to detect the induced emotions (for a review, see Rusting, 1998). In the current experiment, only one of the indirect measures used – word recall – provided statistically significant effects, while the judgment biases in the picture judgment task were too subtle to yield any statistically significant differences between conditions. Physiological indicators of emotions (e.g., skin conductance, heart rate variability, respiration, and facial EMG) could be utilized in future studies to corroborate the experienced emotions (see e.g., Lundqvist, Carlsson, Hilmersson, & Juslin, 2009).

Conclusions

In summary, this study demonstrated that emotionally congruent contextual information about a musical piece has the potential to intensify the emotions induced by that piece – possibly via the *visual imagery* mechanism. This study also showed that narrative descriptions about the original context of a musical piece promoted music-related visual (or narrative) imagery related to those descriptions. This finding could be interpreted as giving support to the notion that music-induced imagery emerges from a narrative mode of listening. It can be concluded that contextual information appears to be a salient and inextricable part of the music listening experience, integrated with musical cues to form a coherent whole.

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Notes

1. Statistical tests indicated that the word-recall variables were non-normally distributed ($p < .05$ using Lilliefors test), but graphical inspection did not reveal any significant skew or other clear signs of non-normality. Nonparametric Wilcoxon rank sum tests produced results that were practically identical to those of the *t* tests, and thus we chose to report the *t* tests as the method is more familiar to readers (and as moderate departures from normality should not be a serious problem when using the *t* test; see, e.g., Sawilowsky & Blair, 1992).
2. The results of these 2-tailed *t* tests were originally published in Vuoskoski and Eerola (2012), and reported here only for reference purposes.
3. This result was originally published in Vuoskoski and Eerola (2012).

References

Baumgartner, T., Esslen, M., & Jancke, L. (2006). From emotion perception to emotion experience: Emotions evoked by pictures and classical music. *International Journal of Psychophysiology*, *60*, 34–43.

Bonde, L. O. (2007). Music as metaphor and analogy: A literature essay. *Nordic Journal of Music Therapy*, *16*, 73–81.

Bouhuys, A. L., Bloem, G. M., & Groothuis, T. G. G. (1995). Induction of depressed and elated mood by music influences the perception of facial emotional expressions in healthy subjects. *Journal of Affective Disorders*, *33*, 215–226.

Bower, G. H. (1981). Mood and memory. *American Psychologist*, *36*, 129–148.

Cone, E. T. (1977). Three ways of reading a detective story – or a Brahms Intermezzo. *The Georgia Review*, *31*, 554–574.

Davis, M. H. (1980). A multidimensional approach to individual differences in empathy. *JSAS Catalog of Selected Documents in Psychology, 10*, 85.

Gabrielsson, A. (2009). The relationship between musical structure and perceived expression. In S. Hallam, I. Cross, & M. Thaut (Eds.), *The Oxford handbook of music psychology* (pp. 141–150). Oxford, UK: Oxford University Press.

Garrido, S., & Schubert, E. (2011). Individual differences in the enjoyment of negative emotion in music: A literature review and experiment. *Music Perception, 28*, 279–296.

Hartig, T., Nyberg, L., Nilsson, L. G., & Gärling, T. (1999). Testing for mood congruent recall with environmentally induced mood. *Journal of environmental psychology, 19*, 353–367.

Isen, A. M. (1985). Asymmetry of happiness and sadness in effects on memory in normal college students: Comment on Hasher, Rose, Zacks, Sanft, and Doren. *Journal of Experimental Psychology: General, 114*, 388–391.

Isen, A. M., & Shalker, T. E. (1982). The effect of feeling state on evaluation of positive, neutral, and negative stimuli: When you 'accentuate the positive', do you 'eliminate the negative'? *Social Psychology Quarterly, 45*, 58–63.

Janata, P. (2004). When music tells a story. *Nature Neuroscience, 7*, 203–204.

Juslin, P. N., Liljeström, S., Västfjäll, D., & Lundqvist, L.-O. (2010). How does music induce emotions? Exploring the underlying mechanisms. In P. N. Juslin & J. A. Sloboda (Eds.), *Handbook of music and emotion: Theory, research, Applications* (pp. 605–642). Oxford, UK: Oxford University Press.

Juslin, P. N., & Västfjäll, D. (2008). Emotional responses to music: The need to consider underlying mechanisms. *Behavioral and Brain Sciences, 31*, 559–575.

Kamen, M. (Composer). (2001). *Band of Brothers* [CD]. New York, NY: Sony Classical Records.

Knobloch-Westerwick, S., Gong, Y., Hagner, H., & Kerbeykian, L. (2012). Tragedy viewers count their blessings: Feeling low on fiction leads to feeling high on life. *Communication Research*. Advance online publication. doi:10.1177/0093650212437758

Koelsch, S., Kasper, E., Sammler, D., Schulze, K., Gunter, T., & Friederici, A. (2004). Music, language, and meaning: Brain signatures of semantic processing. *Nature Neuroscience, 7*, 302–307.

Lavy, M. M. (2001). *Emotion and the experience of listening to music: A framework for empirical research* (Unpublished doctoral dissertation). University of Cambridge, UK.

Lundqvist, L. O., Carlsson, F., Hilmersson, P., & Juslin, P. N. (2009). Emotional responses to music: Experience, expression, and physiology. *Psychology of Music, 37*, 61–90.

Margulis, E. H. (2010). When program notes don't help: Music descriptions and enjoyment. *Psychology of Music, 38*, 285–302.

Maus, F. E. (1997). Narrative, drama, and emotion in instrumental music. *The Journal of Aesthetics and Art Criticism,* 55, 293–303

Newcomb, A. (1984). Once more 'between absolute and program music': Schumann's Second Symphony. *Nineteenth-Century Music, 7*, 233–250.

Orne, M. T. (1962). On the social psychology of the psychological experiment with particular reference to demand characteristics and their implications. *American Psychologist, 17*, 776–783.

Parrott, W. G., & Sabini, J. (1990). Mood and memory under natural conditions: Evidence for mood incongruent recall. *Journal of Personality and Social Psychology, 59*, 321–336.

Phillips, L. H., Bull, R., Adams, E., & Fraser, L. (2002). Positive mood and executive function: Evidence from Stroop and fluency tasks. *Emotion, 2*, 12–22.

Polkinghorne, D. (1988). *Narrative knowing and the human sciences*. Albany: State University of New York Press.

Rusting, C. L. (1998). Personality, mood, and cognitive processing of emotional information: Three conceptual frameworks. *Psychological Bulletin, 124*, 165–196.

Sawilowsky, S. S., & Blair, R. C. (1992). A more realistic look at the robustness and type II error properties of the *t* test to departures from population normality. *Psychological Bulletin, 111*, 352–360.

Shaffer, L. H. (1992). How to interpret music. In M. R. Jones & S. Holleran (Eds.), *Cognitive bases of musical communication* (pp. 263–278). Washington, DC: American Psychological Association.

Tan, S.-L., & Kelly, M. E. (2004). Graphic representations of short musical compositions. *Psychology of Music*, *32*, 191–212.

Thompson, W. F., Graham, P., & Russo, F. A. (2005). Seeing music performance: Visual influences on perception and experience. *Semiotica*, *2005*(156), 203–227.

Thompson, W. F., Russo, F. A., & Quinto, L. (2008). Audio-visual integration of emotional cues in song. *Cognition and Emotion*, *22*, 1457–1470.

Treitler, L. (1989). *Music and the historical imagination*. Cambridge, MA: Harvard University Press.

Vanger, P., Hoenlinger, R., & Haken, H. (1998). Computer aided generation of prototypical facial expressions of emotion. *Methods of Psychological Research Online*, *3*, 25–38.

Vuoskoski, J. K., & Eerola, T. (2012). Can sad music really make you sad? Indirect measures of affective states induced by music and autobiographical memories. *Psychology of Aesthetics, Creativity, and the Arts*, *6*(3), 204–213.

Vuoskoski, J. K., Thompson, W. F., McIlwain, D., & Eerola, T. (2012). Who enjoys listening to sad music and why? *Music Perception*, *29*, 311–317.

Västfjäll, D. (2010). Indirect perceptual, cognitive, and behavioural measures. In P. N. Juslin & J. A. Sloboda (Eds.), *Handbook of music and emotion: Theory, research, applications* (pp. 255–278). Oxford, UK: Oxford University Press.

Watson, D., Clark, L. A., & Tellegen, A. (1988). Development and validation of brief measures of positive and negative affect: The PANAS scales. *Journal of Personality and Social Psychology*, *54*, 1063–1070.

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